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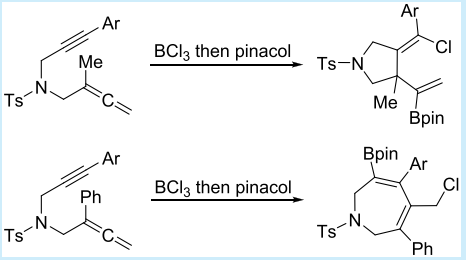
Borylative Cyclization of 1,6-Allenynes Driven by BCl_3

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ABSTRACT: A metal-free intramolecular borylative cyclization of 1,6-allenynes driven by BCl_3 was developed. This method provides a general and practical strategy to construct valuable pyrrolidines containing all-carbon quaternary centers or 3,5-dihydroazepine derivatives depending on the substituents of the allene, with conjugative and sterically hindered phenyl groups favoring the latter.



An attractive goal of organic synthesis is the selective construction of highly functionalized polycyclic molecules, which widely exist in natural products and have a broad range of biological activities, in a step- and atom-economical fashion.[1] Transition-metal-catalyzed cycloisomerization reactions of 1,*n*-allenynes could be regarded as the most efficient method in this regard, as they facilitate the rapid increase in molecular complexity from simple linear substrates.[2] Since Malacria's pioneering work on the cobalt-catalyzed cycloisomerization of 1,6-allenynes,[3] a variety of further-expanded cycloisomerizations have been reported, which proceed through trapping of the intermediates using nucleophilic, oxidative, or reductive reagents.[4] In addition, by combining a C−C bond-forming cyclization with borylation, complex molecules suitable for diverse further functionalizations can be acquired in one step.[5] However, examples that are involved in cycloisomerization of 1,*n*-allenynes to generate borylated molecules are still limited. The Bäckvall group has performed seminal work in this area. In their systematic work, Bäckvall and co-workers found that allenynes can undergo carbocyclization/borylation using a Pd(II) catalyst in the presence of bis(pinacolato)diboron under different conditions to give various boryl-functionalized five-membered rings (Scheme 1a).[5c,6] The Cárdenas group reported a formal hydroborylative carbocyclization of allenynes to obtain allylboronates and alkylboronates catalyzed by $Pd(OAc)_2$ (Scheme 1b).[5n] Although several borylative cycloisomerizations of 1,*n*-allenynes have been disclosed, almost all of these methods suffer from the demand for transition metals, and external additives are usually needed. These requirements have limited the substrate scope and practical applications. Therefore, it is still essential and desirable to develop atom-economical and metal-free methods for the direct preparation of various types of boryl-functionalized polycyclic frameworks, especially those bearing all-carbon quaternary centers, through cycloisomerization of 1,*n*-allenynes.

Recently, the metal-free construction of C−B bonds has attracted interest from the synthetic community.[7] In previous work that is especially related to this study, Erker, Yamaguchi, and co-workers discovered a novel borylative cyclization of 1,2-dialkynylbenzenes using $B(C_6F_5)_3$ and in one example BCl_3 to obtain dibenzopentalenes A and B.[8] Ingleson et al. presented carbocyclization/borylation of a range of diynes to produce boryl-functionalized cyclic compounds using BCl_3.[9] However, to the best of our knowledge, the current studies of metal-free carbocyclization/borylation mainly focus on diynes. It is well-known that allenes share many similar properties with alkynes. It is still challenging but desirable to develop a metal-free method for the construction of borylative all-carbon quaternary stereocenters that would allow a series of derivatizations to access structurally diverse molecules. As part of our work on the formation of C−B bonds, we have for the first time discovered a metal-free carbocyclization/borylation of 1,6-allenynes to afford different borylvinyl pyrrolidines bearing all-carbon quaternary stereocenters as well as 3,5-dihydroazepine derivatives, which are among the challenging-ring-sized skeletons in organic synthesis (Scheme 1c).

In our group, we are interested in extending BCl_3-induced cyclization/borylation to unsaturated compounds, such as alkenes and allenes.[10] We recently accomplished the direct intramolecular aminoboration of sulfonamidoallenes using BCl_3 as the boron source, giving the desired products in

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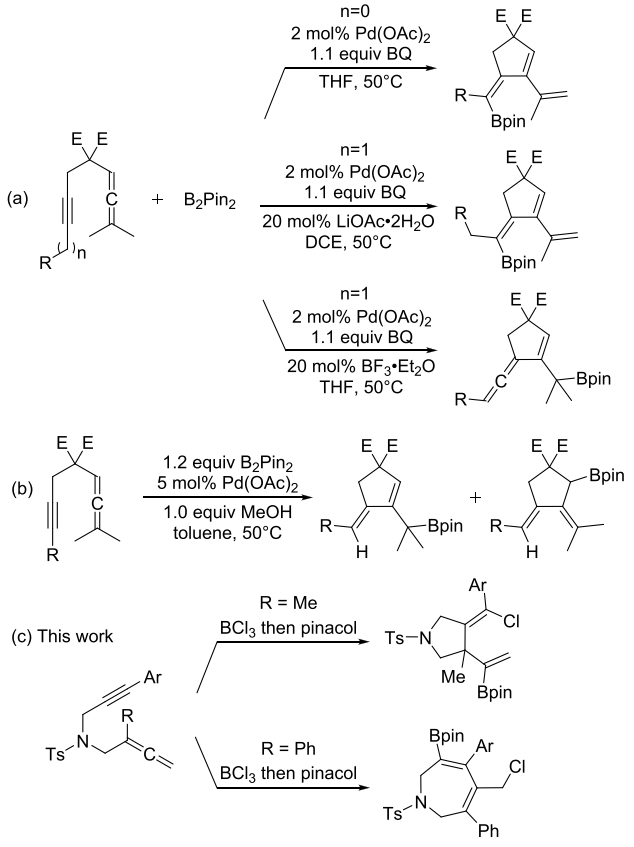


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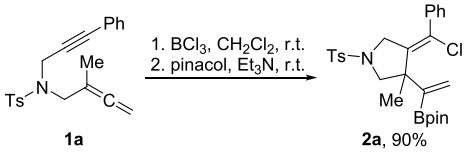
Scheme 1. Borylative Cyclization of 1,*n*-Allenynes



good yields.[10d] Inspired by these results and Ingleson's work,[9] we decided to study the reaction between 1,6-allenynes and BCl_3 to further understand the BCl_3-involved cyclization/borylation reactions on one hand and on the other to expand the scope of the reaction.

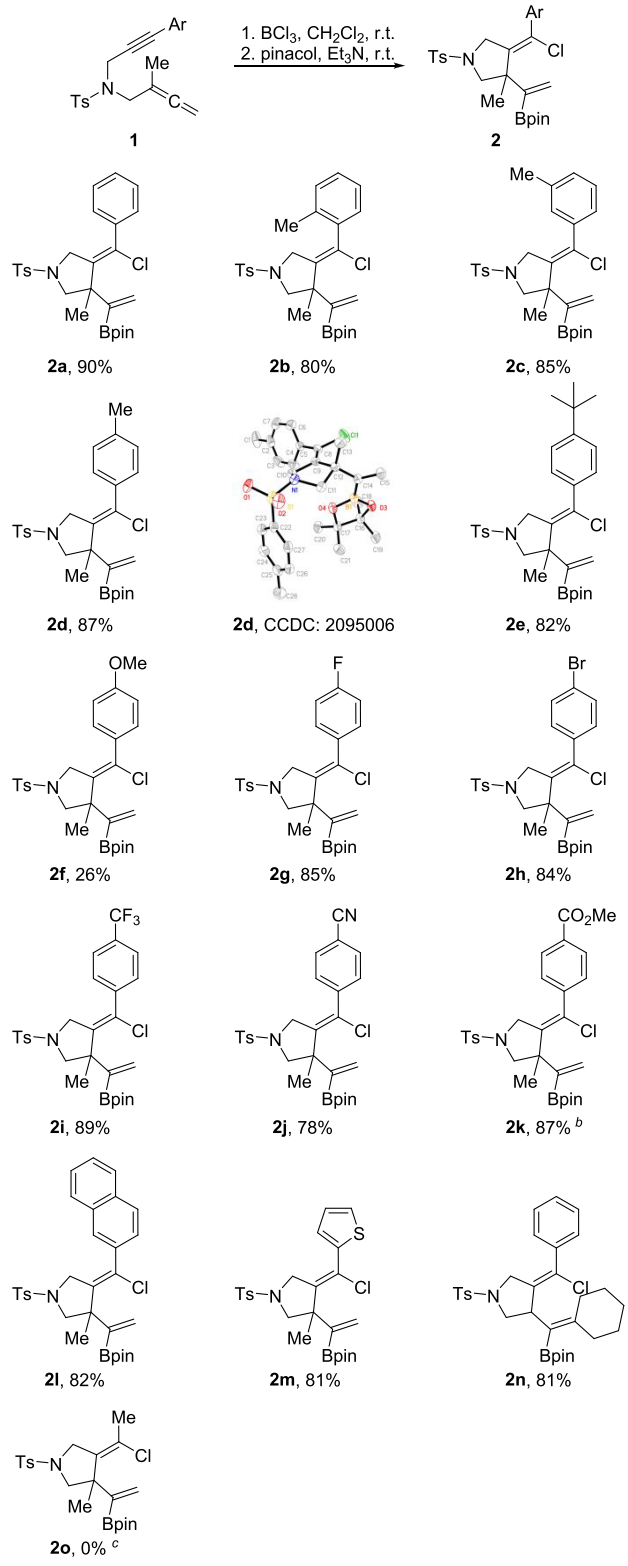
To investigate the borylative cyclization of 1,6-allenynes, initially our study was carried out with 1,6-allenyne **1a** as the model substrate and BCl_3 in a solution of CH_2Cl_2 at ambient temperature for 10 min (Scheme 2). Gratifyingly, treatment with pinacol and Et_3N gave the expected product **2a** in 90% isolated yield, and its structure was confirmed by NMR and HRMS.

Scheme 2. Borylative Cyclization of 1,6-Allenyne 1a



The functional group tolerance of the borylative cyclization proved to be broad (Scheme 3). First, substrates **1b**–**d** bearing a methyl substituent at the *ortho*, *para*, or *meta* position of the benzene ring and *tert*-butyl-substituted 1,6-allenyne **1e** could be subjected to the borylative cyclization smoothly and gave the corresponding products in good yields. The structure of **2d** was also confirmed by single-crystal experiments. In contrast, ether-containing substrate **1f** gave an extremely low yield, which may be ascribed to the cleavage of MeO (as a nucleophile, the electron transfer of MeO-substituted phenyl-

Scheme 3. Reaction Substrate Scope[a]



[a]Reaction conditions: (1) BCl_3 (1.5 equiv); (2) pinacol (1.1 equiv) and Et_3N (15 equiv). [b]2 equiv of BCl_3. [c]No reaction.

acetylene promotes ether cleavage by attack of the anion on $Me^{\delta+}$) with BCl_3, resulting in a series of complex products.

When substrates with weak electron-withdrawing groups such as F (**1g**) and Br (**1h**) on the benzene ring were subjected to the reaction, the desired products were obtained in 84–85%

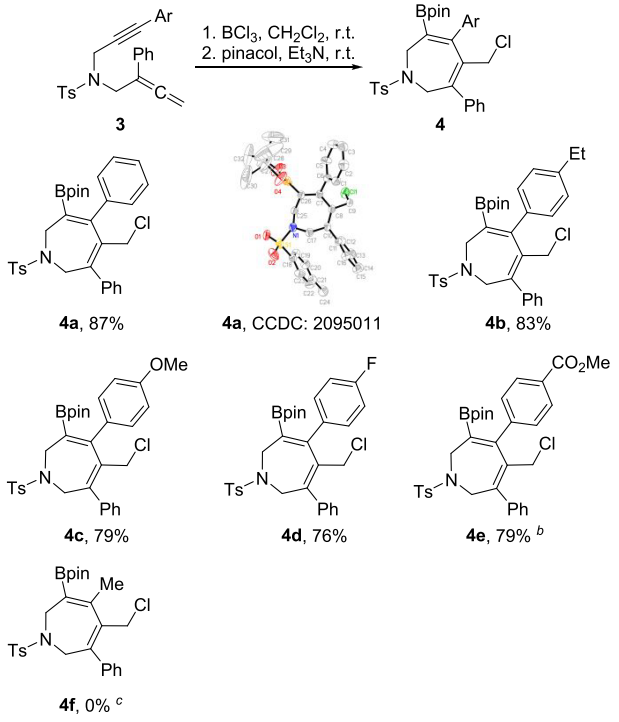
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yield. Moreover, 1,6-allenynes substituted with a relatively stronger electron-withdrawing group such as an ester, CN, or CF_3 group were well-tolerated, providing the desired products in 78−89% yield. For example, substrate **1i** with CF_3 at the *para* position of the benzene ring gave borylative cyclization product **2i** in 89% yield. Substrates containing a Lewis basic nitrile functionality often coordinate with strong boron electrophiles, but in this case CN-containing alkyne **1j** was successfully cyclized and isolated in 78% yield under standard conditions. Ester-containing substrate **1k** was also amenable to cyclization with 2 equiv of BCl_3, which might primarily result from the coordination of the ester moiety with 1 equiv of BCl_3. Moreover, 1,6-allenyne substrates substituted with naphthyl and thienyl groups also underwent the cyclization to form **2l** and **2m**, respectively. When the substrate with trisubstituted allene (**1n**) was subjected to the reaction, the corresponding product was obtained in 81% yield. When alkyl-substituted alkyne **1o** was subjected to the reaction, no product was observed.

Next, reactions of phenyl-substituted 1,6-allenynes were performed to further expand the scope of the reaction. When substrate **3a** was subjected to the reaction conditions, the expected pyrrolidine analogue was not obtained. Treatment with pinacol and Et_3N followed by isolation of the boronate ester derivative enabled confirmation of the structure as the 3,5-dihydroazepine skeleton by single-crystal X-ray diffraction (Scheme 4). Encouraged by this result, we found that substrates bearing electron-neutral (**3a**, **3b**) or electron-donating groups (**3c**) on the benzene ring were well-tolerated, providing the desired products in 79−87% yield, and electron-withdrawing groups such as F (**3d**) and COOMe (**3e**) did not decrease the yields. When alkyl-substituted alkyne **3f** was subjected to the reaction, the substrate disappeared and no desired borylative cyclization product was detected.

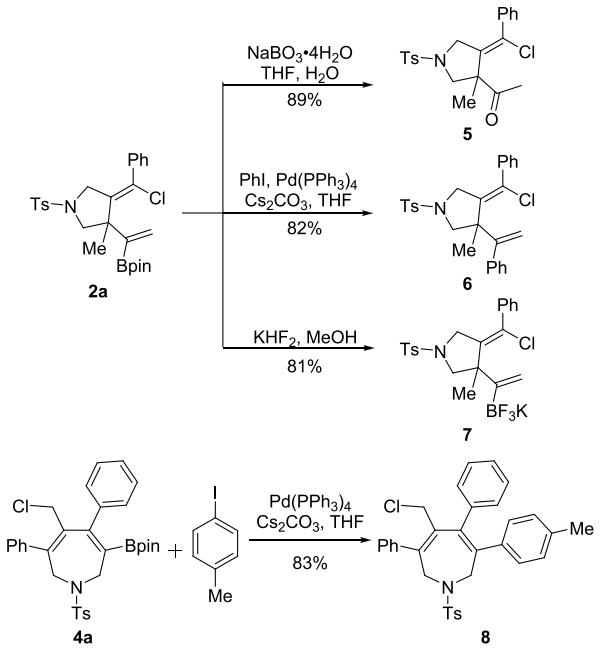
With the aim of demonstrating the application potential of the boronate compounds, functional group transformations were studied with **2a** as the model (see Scheme 5). With

Scheme 4. Borylative Cyclization of Phenyl-Substituted 1,6-Allenynes[a]



[a]Reaction conditions: (1) BCl_3 (1.5 equiv); (2) pinacol (1.1 equiv) and Et_3N (15 equiv). [b]2 equiv of BCl_3. [c]Not detected.

Scheme 5. Derivatization of the Borylative Cyclization Products



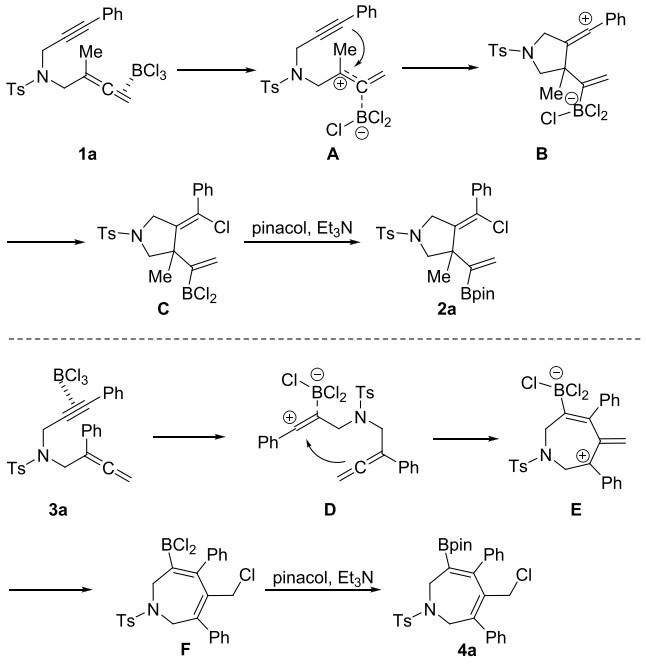
sodium peroxoborate tetrahydrate as an oxidant, the Bpin could be converted to OH, and ketone **5** was obtained as the final product. Suzuki coupling of **2a** with iodobenzene gave the corresponding product **6** in 82% yield. Treating **2a** with aqueous KHF_2 afforded the desired trifluoroborate **7** (Scheme 5). Suzuki coupling of borylative cyclization product **4a** with 1-iodo-4-methylbenzene gave the corresponding product **8** in 83% yields.

According to our calculation results (for details, see the Supporting Information), a relaxed scan of the potential energy surface of BCl_3 approaching substrate **3a** showed that BCl_3 attacks the carbon atom of the alkynyl group more easily. Furthermore, for the phenyl-substituted substrates, electron donor and conjugation effects can reduce the barrier of the seven-membered ring pathway. Once formed, seven-membered-ring products will be produced smoothly, which might result from formation of the stable carbocation. On the other hand, because of steric hindrance by the phenyl group, the five-membered-ring products are not easy to form. When the substituent is methyl, the potential barrier allows only the formation of five-membered-ring products, probably because of the formation of the tertiary carbocation, which is consistent with our experimental results. In addition, the magnitude of the intermediate carbocation may be the key in controlling the products.

The literature results showed that BCl_3 might serve as a carbophilic Lewis acid, resulting in cyclization to zwitterions. On the basis of literature results and the calculation results, a tentative mechanism is proposed and is shown in Scheme 6. Taking **1a** as an example, the complex consisting of substrate **1a** and BCl_3 is formed upon the addition of BCl_3. Nucleophilic

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Scheme 6. Proposed Mechanism



attack on borylvinyl complex **A** along with C−B bond formation can afford zwitterionic vinyl cation intermediate **B**, which is rapidly converted to the corresponding alkenylboron intermediate **C**. With 1,6-allenyne **3a**, the phenyl groups would strongly favor the formation of only seven-membered cyclized intermediate **F**, which is consistent with the DFT calculations (see the Supporting Information for details).

In summary, we have successfully developed the first efficient metal-free intramolecular carbocyclization/borylation of 1,6-allenynes with BCl_3, affording a rapid strategy for the synthesis of a structurally diverse array of borylvinyl-substituted N-heterocycles concomitant with the formation of C−C and C−B bonds in good yields under mild conditions. Importantly, the atom economy and operational simplicity related to this method suggest its great potential for broad applications in the construction of valuable borylvinyl heterocycles in medical chemistry. DFT calculations suggested that different cyclization sites are highly dependent on the conjugative effect and the steric hindrance of substituents.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c03062.

Experimental procedures, compound characterization, NMR spectra, and X-ray diffraction data (PDF)

Accession Codes

CCDC 2095006 and 2095011 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, U.K.; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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